STATISTICS — COMPLETE BEGINNER'S GUIDE

Time Series Regression

From Zero to Full Understanding

> **Who this guide is for.** This document assumes you have never studied statistics before. Every concept is built from the ground up with plain language, visual diagrams, and step-by-step worked examples. Nothing is skipped. If you already have some stats background, the later chapters still add valuable depth and worked examples.

NOTATION GLOSSARY — Read This First

Before anything else, here is every symbol you will encounter, explained in plain English.

Symbol	Name	Plain meaning
Y	Dependent variable	The thing you are trying to predict or explain
X	Independent / explanatory variable	The thing you use to explain Y
t	Time index	Which time period we are in (e.g., t = 1 means year 1)
i	Individual index	Which person/firm/country we are looking at
β (beta)	Regression coefficient	How much Y changes when X changes by 1 unit
α (alpha)	Intercept	The value of Y when all X's = 0
ε (epsilon)	Error / residual	The part of Y that our model fails to explain
ρ (rho)	Autocorrelation coefficient	How strongly this period's error relates to last period's error
γ (gamma)	AR coefficient on lagged Y	How much last period's Y feeds into this period's Y
λ (lambda)	Trend slope	How much the mean of Y grows each period
σ^2	Variance	Measure of spread / unpredictability
Δ (delta)	First difference	Change from one period to the next: ΔX_t = X_t – X_{t–1}
Σ (sigma)	Sum	Add up everything that follows
H_0	Null hypothesis	The "boring" default claim we try to disprove
H_1	Alternative hypothesis	The "interesting" claim we want to support
n	Sample size	Total number of observations
R^2	R-squared	Proportion of Y's variation explained by the model (0 to 1)
SE	Standard error	How much a coefficient estimate typically varies across samples
LM	Lagrange Multiplier	A test statistic we compute to check for autocorrelation
χ^2	Chi-squared	A probability distribution used for some hypothesis tests

Symbol	Name	Plain meaning
DW	Durbin-Watson	A quick test statistic for autocorrelation (limited use)

PART 0 — STATISTICS FOUNDATIONS (Start Here)

Everything in this section is background knowledge that the rest of the guide builds on.

0.1 — What Is Statistics? What Is Regression?

Statistics is the science of learning from data. We observe the world, collect numbers, and try to answer questions like: - Does more education cause higher wages? - Does advertising increase sales? - Does training reduce workplace accidents?

The problem is: the world is messy. Wages are affected by education, *and* experience, *and* gender, *and* luck. Statistics lets us separate these influences.

Regression is the central tool. It finds the straight-line relationship between one or more explanatory variables (X's) and an outcome variable (Y), while controlling for noise.

The Simplest Regression (One Variable)

$$Y = \alpha + \beta X + \varepsilon$$

In plain English:

- **Y** = outcome we want to understand (e.g., hourly wage in euros)
- **X** = thing we think explains Y (e.g., years of education)
- **α** = baseline Y when X = 0 (e.g., wage with zero education)
- **β** = slope — how much Y changes for each 1-unit increase in X
- **ε** = error — everything that moves Y but isn't captured by X

Example: Suppose we find: Wage = 5 + 1.2 × Education + ε

This means: every additional year of education is associated with €1.20/hour more in wages, on average. A person with 10 years of education earns an estimated 5 + 1.2×10 = €17/hour.

The **ε (error)** is what remains unexplained. If the actual wage is €19/hour and our model predicts €17, the residual is ε = 19 − 17 = +2. Our model is off by €2 for that person.

Multiple Regression (Several Variables)

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

Now we control for several things at once. Each β_j tells us: "holding all other X's constant, how much does Y change when X_j increases by 1?" This is the **ceteris paribus** (all else equal) interpretation.

0.2 — How Regression Is Estimated: OLS

OLS = Ordinary Least Squares. It is the standard method for calculating the β values.

The idea: choose the values of α and β that make the errors ε as small as possible across all observations. Specifically, OLS minimises the **sum of squared errors**:

$$\min_{\alpha,\beta} \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (Y_i - \alpha - \beta X_i)^2$$

Why square them? Two reasons: 1. Positive and negative errors don't cancel each other out. 2. Large errors are penalised more than small ones (squaring amplifies big mistakes).

OLS gives us **coefficient estimates** $\hat{\beta}$ ("beta hat") — the best-fitting line through the data.

BLUE: When certain conditions hold (see Section 0.4), OLS is **B**est **L**inear **U**nbiased **E**stimator. "Best" means no other linear method has smaller estimation errors. "Unbiased" means on average the estimates hit the true value (no systematic overestimation or underestimation).

0.3 — R^2 : How Well Does the Model Fit?

$$R^2 = 1 - \frac{\text{Sum of Squared Errors}}{\text{Total Variation in Y}}$$

Range: R^2 is always between 0 and 1.

R^2 value	Meaning
0.0	The model explains nothing — X tells us nothing about Y
0.5	The model explains 50% of Y's variation
1.0	Perfect fit — the model explains everything, zero residual
0.95	Seems great! But in time-series data, this can be a **warning sign** (spurious regression)

Intuition: Imagine Y is the height of ocean waves. Total variation = how much waves differ from their average height. If your model explains 80% of that variation, R^2 = 0.80.

0.4 — Standard Errors, t-statistics, and p-values

Even if OLS gives us $\hat{\beta}$ = 1.2, we cannot be certain the true β is 1.2. If we collected a different sample, we'd get a slightly different estimate. **Standard errors (SE)** measure how much $\hat{\beta}$ varies across samples.

The t-statistic

$$t = \frac{\hat{\beta} - 0}{SE(\hat{\beta})} = \frac{\hat{\beta}}{SE(\hat{\beta})}$$

The t-statistic asks: "Is this estimate far enough from zero to be convincingly non-zero?"

- A **large |t|** (say, $|t| > 2$) means the estimate is large relative to its uncertainty → likely real.
- A **small |t|** (say, $|t| < 1$) means the estimate could easily be zero just by chance → not convincing.

The p-value The **p-value** is the probability of seeing a t-statistic this extreme (or more) if the true β were actually zero.

p-value	Interpretation
$p < 0.01$	Very strong evidence against H_0 (significant at 1%)
$p < 0.05$	Strong evidence against H_0 (significant at 5%) — standard threshold
$p < 0.10$	Weak evidence against H_0 (significant at 10%)
$p > 0.10$	No convincing evidence; cannot reject H_0

Analogy: You flip a coin 20 times and get 17 heads. The p-value asks: "If this were a fair coin (H_0), how likely is getting 17 or more heads?" If that probability is tiny ($p < 0.05$), you conclude the coin is probably not fair — you reject H_0.

Hypothesis Testing in Regression For any coefficient β_j: - **H_0: β_j = 0** — X_j has no effect on Y (null hypothesis) - **H_1: β_j ≠ 0** — X_j does affect Y (alternative hypothesis)

If $p < 0.05$: reject H_0 → X_j is **statistically significant**. If $p \geq 0.05$: fail to reject H_0 → evidence insufficient to confirm an effect.

Important phrasing: We never "accept H_0". We either reject it or fail to reject it. Failing to reject just means the data don't give us enough evidence — not that H_0 is definitely true.

0.5 — The Five Classical Assumptions

OLS is only BLUE when these five conditions hold:

Assumption	What it says	Why it matters
A1	$E(\varepsilon) = 0$	Errors average to zero — no systematic over/under-prediction
A2	$Var(\varepsilon) = \sigma^2$ (constant)	Homoskedasticity — equal spread of errors everywhere
A3	cov(ε_i, ε_j) = 0	Errors are uncorrelated with each other
A4	X's are non-random, not perfectly correlated	We can separate each X's effect
A5	ε ~ Normal (optional)	Needed for exact inference in small samples

When you use time-series data, **A3 is routinely violated** (autocorrelation). This is the central problem this guide addresses. A new assumption is also needed: **stationarity**.

0.6 — Visual Intuition: What a Regression Looks Like

```
Y
|             *
|          *     *
|       *     *
|    *   (fitted line)
|   * *
| *
+---------------------- X
```

Each `*` is a data point. OLS finds the straight line that passes as close as possible to all the points, minimising the total squared vertical distance from points to the line.

The **residuals** (errors) are the vertical distances from each point to the line. Good regression: residuals look like random noise. Bad regression: residuals show a pattern (e.g., all positive on the left, all negative on the right).

PART 1 — TIME SERIES FOUNDATIONS

CHAPTER TS_A — Cautions

What makes time-series data special and why ordinary regression needs modification

A.1 — Cross-Section vs. Time-Series Data

Definitions

Type	Description	Notation	Order matters?
Cross-section	Many units at one point in time	X_i, subscript *i* = individual	No — rows are interchangeable
Time-series	One unit observed repeatedly over time	X_t, subscript *t* = time period	**Yes** — shuffling destroys information
Panel data	Many units over many time periods	X_{it}	Both dimensions matter

Why Order Matters In a cross-section of 1,000 workers you could randomly reorder the rows and your regression results would be identical. In a time series of 40 years of GDP data, row 5 (GDP in year 5) *must* come after row 4 and before row 6. The temporal structure is not a nuisance — it is the entire point.

Visual: What Time-Series Data Looks Like vs. Cross-Section Data **Cross-section data** (each row is a different person):

```
Person | Wage | Education | Age
-------|------|-----------|----
Alice  | 22   |    16     | 34
Bob    | 18   |    12     | 28
Carol  | 35   |    20     | 45
...    | ...  |    ...    | ...
```

You could sort Alice, Bob, Carol in any order. Same result.

Time-series data (each row is a different time period):

```
Year | Belgium GDP | Unemployment | Inflation
-----|-------------|--------------|----------
2000 |   253 bn    |     6.9%     |   2.7%
2001 |   259 bn    |     6.6%     |   2.4%
2002 |   263 bn    |     7.5%     |   1.6%
2003 |   268 bn    |     8.2%     |   1.5%
...  |    ...      |     ...      |   ...
```

Row 2003 MUST come after row 2002. Reordering destroys the story of how the economy evolved.

Example — Hourly Wage

- **Cross-section:** A survey of 500 Belgian employees in 2010, each row one person. You can sort by name, age, sector — the regression of wage on education gives the same answer.
- **Time series:** Average Belgian hourly wage in 1980, 1981, …, 2015. The rows must follow calendar order. Regressing wage_t on time_t captures how wages evolved; shuffling would produce nonsense.

A.2 — Lagged Variables

What a Lag Is A **lag** simply refers to the value of a variable at an earlier time period. Think of it as "looking in the rear-view mirror" — what was the value one period ago? Two periods ago?

If the current period is quarter 3 of 1987:

Notation	Meaning	Value refers to
X_t	Current value of X	Q3 1987
X_{t−1}	One-period lag	Q2 1987
X_{t−2}	Two-period lag	Q1 1987
X_{t−q}	q-period lag	q quarters before Q3 1987

Visual: What a Lag Looks Like in Data

```
Original series X_t:
  t=1: 10
  t=2: 14
  t=3: 11
  t=4: 16
  t=5: 13
```

```
After creating lag X_{t-1}:
  t=1: 10  | X_{t-1}: [missing — no period 0]
  t=2: 14  | X_{t-1}: 10   ← value from t=1
  t=3: 11  | X_{t-1}: 14   ← value from t=2
  t=4: 16  | X_{t-1}: 11   ← value from t=3
  t=5: 13  | X_{t-1}: 16   ← value from t=4
```

Notice: the first row loses its observation because there is no "period 0". Each lag of order q costs you q observations.

Why Lags Are Needed Economic processes rarely respond instantaneously. Consider safety training:

- Hours of safety training are delivered in month t.
- Workers need time to absorb the training and change behaviour.
- The reduction in accidents appears in months t+1, t+2, etc.

A model without lags would miss most of the effect. Lags let us model this delay explicitly.

Another example — central bank interest rates: When a central bank raises interest rates today, the effect on consumer spending might take 6–12 months to fully materialise. Banks need to update loan terms, consumers need to notice higher mortgage costs, businesses need to reassess investment plans. All of this takes time. A model using only current interest rates would dramatically underestimate the total effect.

SPSS Instructions

```
Transform → Create Time Series
  Function: Lag
  Order: 1  → creates X_{t-1}
  Order: 2  → creates X_{t-2}
  (one new variable per order)
Internal variable name: LAG(varname, 1), LAG(varname, 2), …
```

Watch Out: Lost Observations Each lag of order q costs you **q observations** at the start of the series. The lag X_{t−1} is undefined for the first observation (there is no "period zero"). More generally:

- Lag of order 1 → you lose **1** observation (row 1 has no X_{t−1}).
- Lag of order q → you lose **q** observations (rows 1 through q).

So if you start with n = 100 and compute a DL(3) model, your usable sample is only n − 3 = 97. With short time series (n < 50) this can be a serious constraint — each extra lag wastes precious degrees of freedom.

A.3 — Difference Variables

Definition The **first difference** of X_t is the change from one period to the next:

$$\Delta X_t = X_t - X_{t-1}$$

Intuition: If GDP was €100 billion last year and €104 billion this year, the first difference is ΔGDP = 104 – 100 = +€4 billion. It answers: “How much did GDP change?”

This is an **absolute** first difference. For a **seasonal** difference of order 12 (removing year-on-year seasonality in monthly data):

$$\Delta_{12} X_t = X_t - X_{t-12}$$

This says: “How does this January compare to last January?” — removing the seasonal pattern.

Visual: Original vs. Differenced Series

```
GDP level (trending upward — non-stationary):

   GDP |                            *
       |                        * *
       |                    * *
       |                * *
       |            * *
       |        * *
       |    * *
       | * *
       +-------------------------------- time
       Clearly trending → not stationary


ΔGDP (first differences — fluctuates around zero):

  ΔGDP |    *   *       *
       |  *   *   *   *   *   *
       | ———————————————————————  0
       |        *   *       *
       +-------------------------------- time
       Fluctuates around a constant level → stationary
```

This is the visual difference between a series that needs differencing and one that doesn’t.

Interpretation

Original variable	After differencing	Meaning
GDP level (€ billions)	ΔGDP	Change in GDP vs. previous period
log(GDP)	Δlog(GDP) = log(GDP_t / GDP_{t-1})	**Log-return** ≈ percentage growth rate
Monthly unemployment (%)	Δ unemployment	Change in unemployment rate (pp)

Original variable	After differencing	Meaning
log(price)	Δlog(price)	Percentage price change (return)

Why Logarithms? A Beginner's Explanation If a stock price rises from €100 to €200, that is a 100% increase. If another rises from €10 to €20, that is also a 100% increase. But in absolute terms: one rose by €100, the other by €10. They are not comparable in levels.

When you take log(price) and then difference it:

$$\Delta \ln(X_t) = \ln(X_t) - \ln(X_{t-1}) = \ln\left(\frac{X_t}{X_{t-1}}\right) \approx \frac{X_t - X_{t-1}}{X_{t-1}}$$

You get the **percentage change** — which is directly comparable across differently-sized quantities. This is why financial data is almost always analysed in log-returns.

Example: Δln(100 → 102) = ln(102/100) = ln(1.02) ≈ 0.0198 ≈ 2% growth. ✓

SPSS Instructions

```
Transform → Create Time Series
  Function: Difference (absolute)       → ΔX_t = X_t - X_{t-1}
  Function: Percentage change         → relative difference version
Internal name: DIFF(varname, 1), DIFF(varname, 2), …
```

Why Differences Matter

1. **Economic theory talks about changes.** Most macro/micro theories describe how *changes* in X affect *changes* in Y. Differences align your model with theory.
2. **They cure a stochastic trend (unit root).** If X_t is a random walk (X_t = X_{t-1} + ε_t), then ΔX_t = ε_t, which is white noise — stationary. Differencing removes the non-stationarity.
3. **Log-returns are scale-free.** The percentage change in stock price A is directly comparable to the percentage change in stock price B even if they are priced very differently.

A.4 — The Three Problems with Time-Series Regression

Every time you run a regression on time-series data, you must address three specific issues.

Problem 1 — Data Quality and Sample Size Time-series regression is **data-hungry**. Unlike cross-section regression where adding more observations is easy (survey more people), adding more time-series observations means waiting for more years to pass — or finding historical data that may not exist.

Rules of thumb:

Situation	Recommendation
Ideal	n > 100 observations
Minimum	n ≈ 35; at least 10 obs per X-variable in the model

Situation	Recommendation
Yearly data	May need 35+ years; consider monthly or quarterly if available
Monetary variables	Use **real (deflated)** values — inflation creates artificial trends

Why inflation matters: If you regress nominal wages on nominal GDP, both series may trend upward simply because of inflation, not because of any real relationship. Always use inflation- adjusted (real) values for monetary variables.

Example of the inflation problem: Imagine wage = €10/hr in 2000, €20/hr in 2020 (doubled in 20 years). GDP also doubled from €200bn to €400bn. But if *all prices* doubled too (inflation = 100%), nothing actually got better in real terms. The regression would show a "significant positive relationship" between wages and GDP — but it is entirely driven by inflation, not any real economic link. **Use real values.**

Problem 2 — Autocorrelation Classical Assumption A3 states that errors are uncorrelated: $cov(\varepsilon_i, \varepsilon_j) = 0$. In time-series data this is routinely violated. Shocks have persistence — if something unexpected happens today, its effect persists into next period, gradually fading.

Consequences and solutions are covered in **Chapter TS_B**.

Problem 3 — Spurious Regression / Non-Stationarity Imagine regressing Belgian ice cream sales on annual sunspot activity. Both series happen to trend upward over a 30-year window. You would find a hugely significant regression with $R^2 = 0.95$ — even though they have absolutely nothing to do with each other.

The solution is to ensure all variables are stationary before fitting the regression. Details are in **Chapter TS_C**.

CHAPTER TS_B — Autocorrelation

Errors that remember the past: detection with the LMSC test

B.1 — Which Classical Assumption is Violated?

The Five Classical Assumptions (Recap) The OLS estimator is BLUE when the Gauss-Markov conditions hold:

Assumption	Formal statement	Plain English
A1	$E(\varepsilon_t) = 0$	Errors average to zero
A2	$Var(\varepsilon_t) = \sigma^2$ (constant)	Homoskedasticity — errors have equal spread
A3	$cov(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$	Errors are uncorrelated with each other
A4	X's non-random, not perfectly collinear	No perfect multicollinearity
A5 (optional)	$\varepsilon \sim$ Normal	For exact inference in small samples

Autocorrelation violates A3. When errors are correlated over time, $cov(\varepsilon_t, \varepsilon_{t-1}) \neq 0$.

Why A3 Fails So Commonly in Time Series Think of the residual ε_t as "everything that moves Y but isn't in my model." In time-series settings, omitted variables themselves tend to persist over time (institutional inertia, technology cycles, business cycles). Today's omitted shock tends to carry over into tomorrow.

Intuitive example: You model monthly consumer spending using only income. But consumer confidence, which you haven't measured, also drives spending — and consumer confidence is *persistent* (it stays high for several months, then gradually declines). The residuals will therefore form waves: positive for a stretch (when confidence is high), then negative for a stretch (when confidence is low). That is positive autocorrelation.

B.2 — First-Order Autocorrelation

The AR(1) Error Structure We model the autocorrelation in the error term as a first-order autoregressive process:

$$\varepsilon_t = \rho\,\varepsilon_{t-1} + u_t$$

where: - ρ (rho) is the **first-order autocorrelation coefficient** (a number between −1 and 1) - u_t is a **classical white-noise error** — uncorrelated, mean zero, constant variance

White noise means pure randomness — like rolling a fair die each period. No memory, no pattern, no trend. It is the ideal we want our errors to look like.

This equation says: today's error is a fraction ρ of yesterday's error, plus a new fresh shock.

Visual: What Autocorrelated Residuals Look Like **No autocorrelation (ρ = 0) — ideal:**

```
Residuals
  |  *     *        *     *
  |     *     *  *     *     *
0 | ───────────────────────────
  |  *        *  *        *
  |     *  *           *
  +────────────────────────── time
  Random scatter above and below zero. No pattern. A3 satisfied. ✓
```

Positive autocorrelation (ρ > 0) — "waves":

```
Residuals
  |  * * * *
  |*         * *
0 | ───────────────────────────
  |            * * * *
  |                    * * *
  +────────────────────────── time
   Long runs of positive, then long runs of negative. Wave-
like. A3 violated. ✗
```

Negative autocorrelation (ρ < 0) — "zigzag":

```
Residuals
   |  *    *    *    *    *
   |
0  | ────────────────────────────────
   |     *    *    *    *    *
   +──────────────────────────────── time
  Alternates above/below zero every period. Rare in economics. A3 violated. ✗
```

Understanding the Size of ρ

Value of	ρ
$\rho = 0$	No autocorrelation. ε_t = u_t, purely random noise — A3 satisfied.
0 <	ρ
0.3 <	ρ
	ρ
	ρ
	ρ

We require **−1 < ρ < 1** for the error process itself to be stationary and well-behaved.

Understanding the Sign of ρ

Sign	Pattern in residuals	Typical cause in economics
ρ > 0 (positive)	Long runs: + + + + − − − −	Very common. Business cycles, omitted trending variables.
ρ < 0 (negative)	Alternating: + − + − + − …	Rare. Can arise from over-differencing.

Worked analogy for ρ = 0.7: If this quarter's residual is +10, next quarter's residual is expected to be +7 (= 0.7 × 10). The quarter after, +4.9. Then +3.4, then +2.4, then +1.7, and so on — slowly decaying toward zero. A shock takes many periods to fade.

B.3 — Consequences of Autocorrelation

General Case (Static or DL(q) Models) When A3 is violated in a model without lagged Y:

Property	With autocorrelation
OLS estimators $\hat{\beta}$	Still **unbiased** — on average they hit the true β
OLS estimators $\hat{\beta}$	No longer **best** (not BLUE) — a better estimator exists
Standard errors	**Wrong** — they use the wrong formula (one that assumes A3)
t-tests and F-tests	**Invalid** — built on wrong standard errors
Confidence intervals	**Wrong width** — too narrow if $\rho > 0$ (falsely precise)

Property	With autocorrelation
R^2	Unaffected but potentially misleading

The key danger is **false significance**: positive autocorrelation inflates t-statistics, making coefficients appear more significant than they really are.

Analogy for wrong standard errors: Imagine asking 10 people in one household whether they like pizza, and claiming those 10 opinions represent 10 independent data points. They don't — family members influence each other. Your "effective" sample size is much smaller than 10. Autocorrelation does the same thing to time-series data: consecutive observations are not truly independent, so your effective sample size is smaller than it appears, and your standard errors are too small.

Exception: AR(1) Models (Lagged Dependent Variable) When the regression contains Y_{t−1} as a regressor **and** there is autocorrelation in ε:

$$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t \quad \text{with } \mathrm{cov}(\varepsilon_t, \varepsilon_{t-1}) \neq 0$$

The problem is now much worse: the coefficient **estimates themselves become biased**. The reason: Y_{t−1} is a function of ε_{t−1}. If ε_t is correlated with ε_{t−1}, then Y_{t−1} is correlated with ε_t — violating A4. When that happens, OLS is no longer even unbiased.

Practical rule: If you detect autocorrelation in an AR(1) model, re-specify (switch to DL(q)).

B.4 — Detection: The LMSC Test

Step 0 — Graphical Inspection First Always **plot the residuals against time** before running any test. Look for: - **Waves** (long runs above or below zero) → positive autocorrelation, ρ > 0. - **Zigzag** (alternating signs) → negative autocorrelation, ρ < 0. - **Random scatter** (no visible pattern) → A3 plausibly satisfied.

The LMSC Test — Full Procedure **What are we testing?** We want to know if there is a significant relationship between this period's residual and last period's residual. If consecutive residuals are strongly correlated, autocorrelation is present.

Hypotheses: - H_0: ρ = 0 (no autocorrelation — A3 satisfied) - H_1: ρ ≠ 0 (autocorrelation present — A3 violated)

Step 1: Fit the original regression model. Save the residuals e_t.

Step 2: Build the **auxiliary regression**. Regress the residuals on all the original predictors X_{1t}, …, X_{kt} **plus** the lagged residual e_{t−1}:

$$e_t = \alpha_0 + \alpha_1 X_{1t} + \cdots + \alpha_k X_{kt} + \alpha_{k+1} e_{t-1} + u_t$$

Why include the original X's? Without the X's, some of the "relationship" between e_t and e_{t-1} might actually be caused by the X variables themselves (because X's also have temporal structure). Including X's in the auxiliary regression "controls out" their influence, leaving only the pure autocorrelation signal in e_{t-1}.

Think of it as a partial correlation: we want the relationship between consecutive residuals *after accounting for* the effect of the X's.

Step 3: From the auxiliary regression, read R^2 and compute:

$$LM = n \cdot R^2$$

where n is the sample size of the **auxiliary** regression (= n_original − 1).

Why n × R^2? R^2 in the auxiliary regression measures how much of the variation in today's residual is explained by yesterday's residual (and the X's). Multiplying by n scales this into a test statistic with a known distribution.

Step 4: Under H_0, the LM statistic follows $\chi^2(1)$. Compare LM to 3.841 (the 5% critical value for $\chi^2(1)$).

What is $\chi^2(1)$? The chi-squared distribution with 1 degree of freedom is a probability distribution used for test statistics that are squared quantities. The value 3.841 is the threshold: there is only a 5% chance of seeing LM > 3.841 if H_0 is true.

Decision: - If LM > 3.841 (or p-value < 0.05): **Reject H_0** → autocorrelation is present. - If LM ≤ 3.841 (or p-value ≥ 0.05): **Do not reject H_0** → A3 plausibly satisfied.

Higher-Order Tests The procedure above tests for first-order autocorrelation only (e_{t−1} in the auxiliary regression). To test for autocorrelation up to order p, add e_{t−1}, e_{t−2}, …, e_{t−p} to the auxiliary regression. The LM statistic then follows $\chi^2(p)$ under H_0.

Important: The sample size in the auxiliary regression becomes n − p.

Worked Example — Phillips Curve (Australian Quarterly Data) **Original model:** $Y_t = \beta_0 + \beta_1 X_t + \varepsilon_t$ - Y = inflation rate (%), X = change in unemployment rate (pp) - n = 90 quarterly observations (1987Q1 to 2009Q3)

Auxiliary regression results:

Item	Value
n for auxiliary regression	90 − 1 = **89**
R^2 of auxiliary regression	**0.310211**
Coefficient on e_{t−1}	**0.558** (t = 6.219, p < .001)

Computation:

$$LM = n \cdot R^2 = 89 \times 0.310211 = 27.609$$

Conclusion: LM = 27.6 » 3.841. Reject H_0 at any reasonable significance level. Autocorrelation is strongly present. The Phillips curve model is misspecified.

Durbin–Watson as an Alternative SPSS automatically prints the **Durbin–Watson (DW)** statistic.

DW value	Interpretation
Close to **2**	No autocorrelation
Close to **0**	Strong positive autocorrelation (ρ near +1)
Close to **4**	Strong negative autocorrelation (ρ near −1)
Between 1.5 and 2.5	Generally acceptable (rough rule)

Critical limitation: Durbin–Watson is **invalid** whenever the regression contains a lagged dependent variable Y_{t−1}. Always use LMSC as your primary test.

Relationship between DW and ρ:

$$DW \approx 2(1 - \hat{\rho})$$

So DW = 2 when $\hat{\rho} = 0$ (no autocorrelation), DW = 0 when $\hat{\rho} = 1$ (strong positive autocorrelation), and DW = 4 when $\hat{\rho} = -1$ (strong negative autocorrelation).

B.5 — Solutions for Autocorrelation

The Exam Approach: Re-specify the Model The philosophy: autocorrelation in a static model usually indicates **model misspecification**. The error term is carrying memory because dynamics were left out. Adding lags absorbs the dynamic structure and often eliminates autocorrelation.

Re-specification	What you add	Model type
Add lagged X's	X_{t−1}, X_{t−2}, …, X_{t−q}	**DL(q)** — Distributed Lag
Add lagged Y	Y_{t−1}	**AR(1)** — Autoregressive

DL(q) vs. AR(1) — Which to Choose?

Criterion	DL(q) preferred	AR(1) preferred
Residual autocorrelation present	✓ DL is only inefficient, not biased	✗ AR is biased with residual autocorrelation
Multicollinearity concern	✗ DL with many lags is collinear	✓ AR uses only 2 coefficients
Long memory needed	✗ Need many lags = many parameters	✓ $\beta/(1-\gamma)$ captures infinite lag with 2 params
Bottom line	**Safer when autocorrelation is present**	**More parsimonious when A3 satisfied**

Key principle: bias is worse than inefficiency. - DL with autocorrelation → wrong standard errors (bad, but the coefficients are still correct on average). - AR with autocorrelation → wrong coefficient estimates (fundamentally misleading — we don't even know the direction of the effect reliably).

CHAPTER TS_C — Stationarity

Spurious regression, the generalized AR(1), and the Dickey-Fuller test

C.1 — Spurious Regression

The Problem Consider: regress Belgian GDP per capita (1950–2004) on US population density over the same period. You find: - $R^2 \approx 0.96$ (extremely high) - Slope coefficient significant at $p < 0.001$

Yet Belgian GDP and US population density have **no meaningful economic relationship**. This is a **spurious regression** — a statistically convincing result that is economically meaningless.

Why It Happens: Shared Trends Both series happened to grow over the same 54-year period. The regression is picking up their **common time trend**, not any genuine relationship.

Famous real-world example: The number of Nicolas Cage films released per year is significantly correlated with drowning deaths in US swimming pools between 1999 and 2009. Both happened to move similarly over that period — pure coincidence. A regression would give $R^2 \approx 0.67$ and $p < 0.05$. Completely spurious.

The Root Cause: Non-Stationarity The mathematical reason: both variables are **non-stationary** — their means drift over time. Standard regression theory requires stationarity. When it fails, t-statistics don't follow t-distributions, and R^2 doesn't converge to a meaningful value.

Stationarity is therefore a new assumption required for time-series regression.

C.2 — Definition of Stationarity

Formal Definition (Weak/Covariance Stationarity) A time series Y_t is **weakly stationary** if, for every t:

1. **$E(Y_t) = \mu$** — The mean is constant. The series oscillates around a fixed level.
2. **$Var(Y_t) = \sigma^2$** — The variance is constant. Not more volatile over time.
3. **$Cov(Y_t, Y_{t-s}) = \gamma_s$** — The covariance depends only on the gap s, not on when.

Visual: Stationary vs. Non-Stationary Series **Stationary series — fluctuates around a constant level:**

```
Y
  |      *         *    *
  |  *      * *       *    *    *
  | ─────────────────────────────────  ← constant mean
  |   *    *            *      *
  |        *      *
  +───────────────────────────────── time
 Mean is flat. Variance is constant. Keeps returning to the same level.
```

Non-stationary (trending) series — drifts upward:

```
Y
  |                            * *
  |                     * * *
  |              * * *
```

```
|         * * *
| * * *
+------------------------------ time
 Mean is growing over time. Does NOT return to a fixed level.
Running a regression with this series can produce spurious results.
```

Non-stationary (random walk) — wanders unpredictably:

```
Y
  |    *
  |  * * *
  | *     * *
  |          * *
  |             * *
  |               * * * *
  +------------------------------ time
 Neither trends up/down consistently NOR stays near a fixed level.
Goes wherever past shocks have pushed it. Variance grows over time.
```

Intuitive Test: The "Cut in Half" Check Plot the time series. Visually divide it into two halves. If the left half and right half: - Have similar **average levels** → mean is constant → good sign - Have similar **spread/volatility** → variance is constant → good sign - Show similar **patterns** → covariance structure is similar → good sign

If the right half is systematically higher, more volatile, or shows different patterns, you have evidence of non-stationarity.

C.3 — The Generalized AR(1) Model

The Unifying Framework The behaviour of any univariate time series can be studied through the **generalized AR(1)**:

$$Y_t = a + \lambda t + \rho Y_{t-1} + \varepsilon_t$$

where ε_t is white noise: independent draws from N(0, σ^2).

The three parameters determine everything: - **a** — the constant (intercept) - **λ** — the coefficient on time t (slope of deterministic trend) - **ρ** — the autoregressive coefficient (how strongly past Y predicts current Y)

The **critical question** is whether **|ρ| < 1** (stationary family) or **|ρ| = 1** (random walk).

The Four Cases

Case 1 — Basic AR(1): |ρ| < 1, a = 0, λ = 0

Model: Y_t = ρ Y_{t-1} + ε_t

Intuition: Each period, Y reverts toward zero. If Y is currently +10, next period it will tend to be ρ×10 (closer to zero). The series has a "gravitational pull" back to zero.

By substituting the equation for Y_{t−1} back into Y_t, then Y_{t−2} back in, and so on:

$$Y_t = \varepsilon_t + \rho\varepsilon_{t-1} + \rho^2\varepsilon_{t-2} + \cdots + \rho^t Y_0$$

This is an **infinite moving average** of all past shocks, with geometrically declining weights. Because |ρ| < 1, the weights ρ, ρ^2, ρ^3, … shrink toward zero — old shocks become irrelevant.

Properties: | Property | Formula | Stationarity check | |———-|———|—————————-| | Mean | E(Y_t) ≈ 0 | ✓ Constant | | Variance | Var(Y_t) = $\sigma^2/(1-\rho^2)$ | ✓ Constant (requires |ρ| < 1) | | Covariance | Cov(Y_{t−s}, Y_t) = $\sigma^2\rho^s/(1-\rho^2)$ | ✓ Depends only on lag s | | Autocorrelation | r(Y_{t−s}, Y_t) = ρ^s | Decays exponentially to 0 |

✓ STATIONARY

Example: Y_t = 0.7 Y_{t−1} + ε_t. A shock of +10 in period t: period t+1: +7.0, t+2: +4.9, t+3: +3.4, t+4: +2.4, …, eventually → 0.

Case 2 — AR(1) with constant: |ρ| < 1, a ≠ 0, λ = 0

Model: Y_t = a + ρ Y_{t-1} + ε_t

Property	Formula
Mean	μ = a/(1−ρ)
Variance	$\sigma^2/(1-\rho^2)$

✓ STATIONARY — but now around a non-zero mean μ = a/(1−ρ).

Intuition for the mean formula: In long-run equilibrium, Y_t = Y_{t−1} = μ (no expected change). Substituting: μ = a + ρμ → μ(1−ρ) = a → μ = a/(1−ρ).

Example: Y_t = 5 + 0.6 Y_{t−1} + ε_t. Long-run mean = 5/(1−0.6) = 12.5. The series oscillates around 12.5, never drifting away permanently.

Case 3 — AR(1) with deterministic trend: |ρ| < 1, a ≠ 0, λ ≠ 0

Model: Y_t = a + λt + ρ Y_{t-1} + ε_t

The λt term makes the mean grow linearly over time. The series **is not stationary** (the mean changes with t). However, the **deviation from the trend line** is stationary.

Visual:

```
Y  |                  *  ← actual data
   |               * /
   |            * / trend line (growing by λ per period)
   |         * /
   |      * /
   |   * /
```

```
+──────────────────── time
Deviations from the trend line (the gaps) ARE stationary.
The trend line itself is NOT stationary.
```

✗ **NOT STATIONARY** in levels, but ✓ **stationary about a deterministic trend**.

Fix: Add t as a regressor in the regression (see C.5, Fix A).

Case 4 — Random Walk: $|\rho| = 1$

When $\rho = 1$, $Y_t = a + Y_{t-1} + \varepsilon_t$.

Crucial difference from Case 1: Each shock is permanent. By repeated substitution: $Y_t = Y_0 + \varepsilon_1 + \varepsilon_2 + \ldots + \varepsilon_t$

The series is the accumulated sum of ALL past shocks. None of them ever fade.

Sub-case	Model	Mean behaviour	Variance behaviour
Pure random walk (a=0, λ=0)	$Y_t = Y_{t-1} + \varepsilon_t$	$E(Y_t) = Y_0$ (constant)	$Var(Y_t) = t\sigma^2$ → **grows without bound**
Random walk with drift (a≠0, λ=0)	$Y_t = a + Y_{t-1} + \varepsilon_t$	$E(Y_t) = ta + Y_0$ → **drifts**	Also grows with t

Why the variance explodes: $Var(Y_t) = Var(\varepsilon_1 + \varepsilon_2 + \ldots + \varepsilon_t) = Var(\varepsilon_1) + Var(\varepsilon_2) + \ldots + Var(\varepsilon_t) = t\sigma^2$

Each period adds another independent shock. Unlike Case 1 where past shocks had weights ρ^s that decay, here all past shocks carry full weight forever.

Visual: Random walk vs. AR(1)

```
AR(1) with ρ = 0.7 — stationary:      Random walk (ρ = 1) — non-
stationary:
Y                                      Y
  |  *   *                              |               * *
  |*  * *  * *     *                    |          * * *
  |─────────────────────                |       * *
  |      *    * *  * *                  |    * *
  |           *                         | *
  +────────────────────── time          +────────────────────── time
 Keeps returning to mean = 0          Wanders freely, never returns
 Shocks fade out                        Shocks accumulate forever
```

✗ **ALL NON-STATIONARY.** Fix: Take first differences.

Summary Card

Case	ρ	λ	Stationary?	Fix
Basic AR(1)		ρ	<1	0

Case	ρ	λ	Stationary?	Fix
AR(1) + constant		ρ	<1	0
AR(1) + deterministic trend		ρ	<1	$\neq 0$
Random walk		ρ	=1	any

C.4 — Deterministic vs. Stochastic Trends

This distinction is crucial because the two types of non-stationarity require **different fixes**. Applying the wrong fix can make things worse.

Feature	Deterministic Trend	Stochastic Trend (Random Walk)
Mathematical source	λt term in the model	
Mean	Changes predictably (grows by λ per period)	May drift, but unpredictably
Variance	**Constant**	**Grows with time**
Shocks	Temporary — series returns to trend after a shock	Permanent — series never recovers
Example	GDP trend growth of 2% per year	S&P 500 stock price index
Correct fix	**Detrend** or add t as regressor	**First difference**
Wrong fix	First-differencing over-differences → negative autocorrelation	Detrending doesn't remove stochastic trend

Key intuition — permanence of shocks: - **Deterministic trend:** A recession pushes GDP below its trend line, but in subsequent years GDP returns to the same trend. The shock is temporary. - **Random walk:** A recession permanently lowers the level of the series. Future values are all shifted down by the same amount. The shock is permanent and the series has no "memory" of where it "should" be.

C.5 — Solutions for Non-Stationarity

Fix A — Deterministic Trend ($|\rho| < 1, \lambda \neq 0$) **Option 1 — Detrending:** Estimate the trend (regress Y_t on t) and subtract it: $\tilde{Y}_t = Y_t - \hat{\mu} - \hat{\delta}t$ (the residuals from this auxiliary regression) Then use $\tilde{Y}_t$ in place of Y_t in all subsequent analysis.

Option 2 — Model the trend explicitly (preferred in practice): Include t as an additional regressor in your main model:

$$Y_t = \beta_0 + \beta_1 X_t + \lambda t + \varepsilon_t$$

This is equivalent to Option 1 but simpler — you let OLS handle the detrending simultaneously. The coefficient on t absorbs the linear trend; the coefficient on X_t now measures the relationship *after controlling for the common time trend*.

Option 3 — Seasonal differencing: For data with seasonality (monthly, quarterly), use: Δ_{12}X_t = X_t − X_{t−12}

This removes any pattern that repeats annually.

Fix B — Stochastic Trend / Unit Root (|ρ| = 1) **Take first differences.** This is the only valid fix for a random walk.

$$\Delta Y_t = Y_t - Y_{t-1}, \quad \Delta X_t = X_t - X_{t-1}$$

Why it works: If Y_t = Y_{t-1} + ε_t, then: ΔY_t = Y_t − Y_{t-1} = ε_t ← white noise, stationary ✓

The difference operation strips away the accumulated shock history, leaving only the new shock.

How many times to difference? - If Y_t has one unit root → take **first differences** ΔY_t - If ΔY_t still has a unit root → take **second differences** Δ^2Y_t = ΔY_t − ΔY_{t−1} - Most macroeconomic series need only one round of differencing.

> **Important caveat** — **Cointegration:** There is one exception to "always difference I(1) series." If two non-stationary series are **cointegrated** — meaning they are genuinely tied together by an economic relationship and their *difference* is stationary — you should NOT difference them. Instead, run the regression in levels (the "cointegrating regression"). This preserves the long-run relationship. If you difference cointegrated variables, you throw away the most economically meaningful information. *Detecting cointegration is beyond this course's scope, but always consider whether two trending series might be genuinely linked before differencing.*

C.6 — The Dickey-Fuller Test

Purpose The Dickey-Fuller test is a **formal unit root test**: it decides whether ρ = 1 (non-stationary) or |ρ| < 1 (stationary). It converts a visual judgment into a rigorous hypothesis test.

Hypotheses

- **H_0: ρ = 1** — the series has a unit root, is NOT stationary
- **H_1: |ρ| < 1** — the series IS stationary (possibly about a deterministic trend)

Note the asymmetry: H_0 is the dangerous case (non-stationarity). We want to reject H_0. **Failing to reject H_0 means we cannot prove stationarity** — assume the series needs differencing.

The Reparameterisation Trick Instead of testing H_0: ρ = 1 directly, subtract Y_{t-1} from both sides:

$$\Delta Y_t = a + \lambda t + \alpha_1 Y_{t-1} + \varepsilon_t$$

where **α_1 = ρ − 1**. Now: - H_0: ρ = 1 ⟺ **α_1 = 0** - H_1: |ρ| < 1 ⟺ **α_1 < 0** (must be negative since ρ < 1 means ρ−1 < 0)

We test whether α_1 is significantly **negative** — a one-sided test.

Why reparameterise? OLS can easily test whether a coefficient equals zero. Testing "does ρ = 1" directly is mathematically awkward. By defining α_1 = ρ − 1, we convert the question into "does α_1 = 0?" which is standard.

Why the Normal t-Table Cannot Be Used Here This is a critical point that confuses many students.

Normally, when we test whether a regression coefficient is zero, the t-statistic follows a t-distribution (or normal distribution in large samples). We use critical values like ±1.96 (at 5%) or ±2.58 (at 1%).

Under the null hypothesis of a unit root (H_0: $\rho = 1$), the standard statistical theory completely breaks down. Here is the intuitive reason:

The standard theory assumes the data are stationary. Under H_0, Y_t is a random walk — it is *not* stationary. Its variance grows over time. When we use Y_{t-1} as a predictor of ΔY_t, we are using an increasingly "large" (non-stationary) variable. The usual asymptotic results that justify the t-distribution do not apply.

The consequence: the actual distribution of the t-statistic under H_0 is **shifted to the left** (more negative) compared to the usual t-distribution. If you used standard critical values (like −1.96 for 5%), you would almost never reject H_0 — even when the series is clearly stationary. You would incorrectly conclude "unit root present" for most stationary series.

This is why Dickey and Fuller derived new, special critical values through simulation:

DF Version	1% critical value	5% critical value	10% critical value
Version 1 (no trend)	−3.43	**−2.86**	−2.57
Version 2 (with trend)	−3.96	**−3.41**	−3.13

These are more negative than the usual −1.96, reflecting the leftward shift of the distribution.

Decision rule: Reject H_0 (conclude stationarity) if and only if the t-statistic on Y_{t−1} is **more negative** than the critical value.

Example: t-stat = −3.10. - Version 1 at 5%: −3.10 < −2.86 → Reject H_0 → stationary ✓ - Version 2 at 5%: −3.10 > −3.41 → Fail to reject → unit root ✗

The choice of version matters!

The Two Versions

Version	Regression	When to use
Version 1 — no trend	$\Delta Y\_t = \alpha_0 + \alpha_1 Y\_\{t-1\} + \varepsilon\_t$	No visible trend in time plot
Version 2 — with trend	$\Delta Y\_t = \alpha_0 + \lambda t + \alpha_1 Y\_\{t-1\} + \varepsilon\_t$	Clear trend in time plot, or when unsure

Version 2 is the **safer default**: if a trend is present but you use Version 1, your test is misspecified. If no trend is present but you use Version 2, you merely lose one degree of freedom.

Procedure (4 Steps)

1. **Plot Y_t against time.** Does it trend? Choose Version 1 (no trend) or Version 2 (trend or unsure).
2. **Create ΔY_t** (first difference of Y) and Y_{t−1} (one lag of Y).
3. **Run the chosen DF regression** (ΔY_t on Y_{t−1}, plus t for Version 2).
4. **Find the t-statistic for the coefficient of Y_{t−1}**. Compare to adapted critical values.

What to Do After the DF Test

DF result	Action
Reject H_0 (stationary)	Use the series **in levels** in your model
Fail to reject H_0 (unit root)	Take **first differences**, then re-run DF on ΔY_t
DF on ΔY_t: Reject H_0	ΔY_t is stationary; use **first differences** in the model
DF on ΔY_t: Still fail to reject	Rare. Consider second differences

---

Reference — Mean of a Stationary AR(1)

Model	Stationarity	Long-run mean
Y_t = ρ Y_{t-1} + ε_t	✓ Stationary	μ = 0
Y_t = a + ρ Y_{t-1} + ε_t	✓ Stationary	μ = a/(1−ρ)
Y_t = a + λt + ρ Y_{t-1} + ε_t	✗ Not stationary	Stationary about μ + δt

---

CHAPTER TS_D — Dynamic Models

DL(q), AR(1), total multipliers, and the time-series exam workflow

D.1 — Distributed Lag Model DL(q)

Motivation A **static model** assumes the entire effect of X on Y is instantaneous:

$$Y_t = \alpha + \beta_0 X_t + \varepsilon_t$$

But most economic effects have **delayed responses**: - A tax cut today may not fully boost consumption until next quarter. - Training employees today reduces accidents over the next several months. - An interest rate rise starts affecting mortgage payments immediately but only slows the housing market after several months.

The **Distributed Lag model of order q** captures these delays by including the current and q lagged values of X:

$$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \beta_2 X_{t-2} + \cdots + \beta_q X_{t-q} + \varepsilon_t$$

The effect of X is **distributed across** q+1 periods — hence the name.

Visual: How a DL(q) Model Distributes the Effect Over Time Suppose safety training reduces accidents. Here is how the DL(2) model (q=2) sees it:

```
Training hours in month t
          |
          |─── β₀ ───> Effect on accidents in month t   (immediate)
        |─── β₁ ───> Effect on accidents in month t+1 (1-period lag)
        |─── β₂ ───> Effect on accidents in month t+2 (2-period lag)

Total effect of one training hour = β₀ + β₁ + β₂  (the total multiplier)
```

The coefficients β_0, β_1, β_2 can all be different magnitudes — the effect profile is completely flexible.

Interpretation of Coefficients Each β_k is interpreted *ceteris paribus* — holding all other X variables constant.

β_k = the average change in Y_t when X_{t−k} rises by 1 unit, all other lagged values held constant.

The most practically useful comparison is between a **temporary** and a **maintained** change:

Type of change	Effect at lag k	Long-run total
Temporary (X rises by 1, then returns)	β_k is the additional effect k periods later	Ends after lag q
Maintained (X rises by 1 permanently)	Cumulative: $\beta_0 + \beta_1 + \ldots +$ β_k after k periods	**Total multiplier** = Σβ_k

The Total Multiplier

$$\text{Total multiplier} = \sum_{k=0}^{q} \beta_k = \beta_0 + \beta_1 + \cdots + \beta_q$$

Interpretation: The total multiplier is the **long-run effect on Y of a permanent 1-unit increase in X**. It answers: "If X goes up by 1 unit and stays there forever, by how much will Y eventually change?"

D.2 — Choosing the Optimal Lag Length q

Why Not Just Include All Lags? Including too many lags creates problems: 1. **Lost observations:** each lag costs one more observation. 2. **Multicollinearity:** X_t, X_{t-1}, X_{t-2} are typically very similar — their correlation is high. This inflates standard errors for all coefficients. 3. **Overfitting:** unnecessary lags soak up degrees of freedom without improving model quality.

The Top-Down Procedure Start from a maximum lag q_max and progressively drop the highest lag that is not significant:

1. Fit DL(q_max). Check the t-test for β_{q_max}.
2. If β_{q_max} is **not significant**: drop lag q_max, refit DL(q_max − 1).
3. Repeat until the highest remaining lag is **significant**.
4. That is your **optimal lag length q**.

Important: Once you find the optimal q, do **not** drop intermediate insignificant lags. If β_2 is significant but β_1 is not, you still keep both — you cannot have lag 2 without lag 1. This would create a gap in the lag structure with no theoretical justification.

Worked Example — Safety Training and Accident Losses Goal: model monthly accident losses (Y, euros) as a function of safety training hours (X), with q_max = 4.

Step	Model fitted	Result	Action
1	DL(4)	β_4: t = 0.83, p = 0.41	Not significant → Drop lag 4
2	DL(3)	β_3: t = 1.21, p = 0.23	Not significant → Drop lag 3
3	DL(2)	β_2: t = 1.56, p = 0.12	Not significant → Drop lag 2
4	DL(1)	β_1: t = −2.87, p = 0.005	**Significant → Stop. Optimal q = 1**

The final model is: Ŷ_t = α̂ + $\hat{\beta}_0$ X_t + $\hat{\beta}_1$ X_{t−1}

Report: - **Immediate effect:** $\hat{\beta}_0$ - **One-period lagged effect:** $\hat{\beta}_1$ - **Total multiplier:** $\hat{\beta}_0 + \hat{\beta}_1$

D.3 — Autoregressive Model AR(1)

From Distributed Lags to Autoregression Instead of explaining today's Y by past values of X, the AR(1) model lets **past Y** carry the memory of the system:

$$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$$

Why include Y_{t−1}? Because in many economic processes, the current state of Y is the best single predictor of the next state. Consumer spending this quarter depends heavily on consumer spending last quarter (habits, contracts, inertia). By including Y_{t−1}, we capture all persistence effects in a single coefficient γ.

Interpreting the Coefficients

Coefficient	Interpretation
α	Intercept
β	The **immediate effect**: a 1-unit rise in X_t causes Y_t to rise by β, holding Y_{t−1} constant
γ	The **persistence coefficient**: how much of last period's Y carries forward. Requires
β/(1−γ)	The **total multiplier**: long-run effect of a permanent 1-unit rise in X

The Total Multiplier Formula

$$\text{Total multiplier (AR1)} = \frac{\beta}{1-\gamma}$$

This requires **|γ| < 1** (stationarity condition).

Proof sketch: In long-run equilibrium, Y_t = Y_{t−1} = Y* (steady state): Y* = α + β X* + γ Y* → Y*(1−γ)* = *α + β X* → Y* = α/(1−γ) + [β/(1−γ)] X*

So a 1-unit permanent increase in X* raises long-run Y* by β/(1−γ).

Dynamic Path After a Temporary Shock Suppose X rises by 1 unit only in period t, then returns to its original value:

Period	Additional effect on Y (above baseline)
t	β
t+1	γ · β
t+2	$\gamma^2 \cdot \beta$
t+3	$\gamma^3 \cdot \beta$
t+k	γ^k · β

The effect decays geometrically at rate γ.

Worked Example — Education Spending and GDP Growth **Data:** Yearly US data since 1910. - X_t = education spending per child (dollars) - Y_t = GDP growth rate (%)

Fitted AR(1) model: Ŷ_t = 1.01 + 0.009 X_t + 0.627 Y_{t−1}

Component	Value	Meaning
α̂	1.01	Baseline growth when X = 0 and Y_{t−1} = 0
β̂	0.009	A $1 increase raises GDP growth by 0.009 pp immediately
γ̂	0.627	62.7% of last year's GDP growth persists into this year
Total multiplier	0.009/(1–0.627) = **0.024**	A permanent $1/child increase raises long-run GDP growth by 0.024 pp

---

D.4 — Link Between AR(1) and DL(∞)

AR(1) Is Secretly an Infinite Distributed Lag We can show that AR(1) is equivalent to a DL(∞) with geometrically declining coefficients. Substitute Y_{t–1} into the AR(1) equation, then Y_{t–2}, and continue indefinitely:

$$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$$

Substituting once (Y_{t-1} = α + β X_{t-1} + γ Y_{t-2} + ε_{t-1}):

$$= \alpha(1+\gamma) + \beta X_t + \beta\gamma X_{t-1} + \gamma^2 Y_{t-2} + \gamma\varepsilon_{t-1} + \varepsilon_t$$

Continuing to infinity:

$$Y_t = \frac{\alpha}{1-\gamma} + \beta X_t + \beta\gamma X_{t-1} + \beta\gamma^2 X_{t-2} + \beta\gamma^3 X_{t-3} + \cdots$$

This is a DL(∞) model with **geometrically declining lag coefficients**: β, $\beta\gamma$, $\beta\gamma^2$, $\beta\gamma^3$, …

Total Multiplier from the Geometric Series

$$\text{Total multiplier} = \sum_{k=0}^{\infty} \beta\gamma^k = \beta \cdot \frac{1}{1-\gamma} = \frac{\beta}{1-\gamma}$$

Why does $\Sigma\gamma^k = 1/(1-\gamma)$? This is the geometric series formula. Here is the intuition:

Let $S = 1 + \gamma + \gamma^2 + \gamma^3 + \ldots$ (infinite sum, with $|\gamma| < 1$) Then $\gamma S = \gamma + \gamma^2 + \gamma^3 + \gamma^4 + \ldots$ (multiply both sides by γ) Subtract: $S - \gamma S = 1$ (all terms cancel except the first) So $S(1 - \gamma) = 1$, which gives $S = 1/(1-\gamma)$. ✓

This only works when $|\gamma| < 1$, so the terms shrink toward zero and the sum converges. If $|\gamma| \geq 1$, the terms grow (or stay the same), the sum is infinite, and the total multiplier is undefined — which is why stationarity ($|\gamma| < 1$) is required.

AR(1) vs. DL(q): A Full Comparison

Feature	DL(q)	AR(1)
Lag structure	q+1 free coefficients β_0, β_1, …, β_q	Geometric decay — only β and γ
Parameters needed	q+1	2
Multicollinearity	High for large q	Low (only one lag of Y needed)
Bias when A3 violated	Unbiased (only inefficient)	**Biased**
Lag structure flexibility	Completely flexible	Constrained to geometric decay
When to prefer	When autocorrelation is suspected	When parsimony matters and A3 is satisfied

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D.5 — The Three-Step Exam Workflow

Step 1 — Stationarity Check (Chapter TS_C) For **each** variable in the model:

1. Plot the series against time. Does it trend up/down? Does variance change?
2. Choose DF version:

    - **No trend visible** → Dickey-Fuller Version 1
    - **Clear trend** or **unsure** → Dickey-Fuller Version 2
3. Run the DF regression, read the t-statistic for the coefficient on Y_{t−1}.
4. Compare to adapted critical values (−2.86 for V1 at 5%, −3.41 for V2 at 5%).
5. **Reject H_0** → series is stationary → use in levels.
6. **Fail to reject H_0** → unit root → take first difference, then re-run DF on ΔY_t.

Step 2 — Fit the Model (Chapter TS_D) **For DL(q):** - Start with q = q_max (given in the question). - Drop highest non-significant lag; repeat until highest lag is significant. - Report all coefficients and the total multiplier = Σβ_k.

For AR(1): - Fit Y_t = α + β X_t + γ Y_{t-1} + ε_t directly. - Report β, γ, and total multiplier = β/(1−γ).

Step 3 — Assumption Check with LMSC (Chapter TS_B)

1. **State hypotheses:** H_0: ρ = 0 (no autocorrelation) vs. H_1: ρ ≠ 0.
2. **Write the auxiliary regression:** e_t = α_0 + α_1 X_{1t} + … + α_k X_{kt} + α_{k+1} e_{t−1} + u_t
3. **Compute:** LM = n_aux × R^2_aux ~ $\chi^2(1)$ under H_0
4. **Compare to** $\chi^2(1, 5\%) = 3.841$.
5. **Conclude** with a full sentence.

D.5 (Worked) — Consumption and Income (Log-Log Model)

Setup: Both income (X) and consumption (Y) are log-transformed. Fit a DL(q) on first differences (log-returns). Data: n = 162 quarterly observations.

Step 1 — Stationarity Both ln(Y) and ln(X) show clear upward trends.

Variable	DF Version	t-statistic	Critical (5%)	Conclusion
ln(Y) — level	V2 (trend)	−1.82	−3.41	Fail to reject → unit root
ln(X) — level	V2 (trend)	−1.65	−3.41	Fail to reject → unit root
Δln(Y) — difference	V1 (no trend)	**−9.95**	−2.86	**Reject → stationary ✓**
Δln(X) — difference	V1 (no trend)	**−12.72**	−2.86	**Reject → stationary ✓**

Both variables need first differencing.

Step 2 — Fit DL(q), q_max = 3

Model	Highest lag t-stat	Significant?	Action
DL(3)	β_3: t = 0.91	No	Drop lag 3
DL(2)	β_2: t = 1.38	No	Drop lag 2

Model	Highest lag t-stat	Significant?	Action
DL(1)	**β_1: t = 2.74**	**Yes**	**Stop. q = 1**

Fitted model: $\Delta\hat{Y}_t = 0.004 + 0.350\,\Delta X_t + 0.182\,\Delta X_{t-1}$

Coefficient	Value	Interpretation
$\hat{\beta}_0$	0.350	A 1 pp rise in income growth raises consumption growth by 0.350 pp this quarter
$\hat{\beta}_1$	0.182	A 1 pp rise last quarter's income growth raises this quarter's consumption by 0.182 pp
Total multiplier	**0.532**	A permanent 1 pp rise in income growth raises consumption growth by 0.532 pp long-run

Step 3 — LMSC Test Auxiliary regression: $e_t = \alpha_0 + \alpha_1\,\Delta X_t + \alpha_2\,\Delta X_{t-1} + \alpha_3\,e_{t-1} + u_t$ $n_{aux} = 161$, $R^2_{aux} = 0.002618$

LM = 161 × 0.002618 = **0.4215** p-value = $P(\chi^2(1) \geq 0.4215)$ = **0.5162**

Conclusion: Fail to reject H_0. No significant autocorrelation detected. A3 plausibly satisfied. ✓

FORMULA SHEET — The Four Equations You Must Know

1. DL(q) Model

$$Y_t = \alpha + \beta_0 X_t + \beta_1 X_{t-1} + \cdots + \beta_q X_{t-q} + \varepsilon_t$$

Total multiplier = $\beta_0 + \beta_1 + \ldots + \beta_q$ = $\boldsymbol{\Sigma\beta_k}$

2. AR(1) Model

$$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + \varepsilon_t$$

Total multiplier = $\boldsymbol{\beta/(1-\gamma)}$ (requires $|\gamma| < 1$)

3. LMSC Test

Auxiliary regression: $e_t = \alpha_0 + \alpha_1 X_{1t} + \ldots + \alpha_k X_{kt} + \alpha_{k+1} e_{t-1} + u_t$

$$LM = n \cdot R^2 \sim \chi^2(1) \text{ under } H_0$$

H_0: $\rho = 0$ (no autocorrelation) | H_1: $\rho \neq 0$ (autocorrelation present) Critical value at 5%: $\chi^2(1) = 3.841$

4. Dickey-Fuller Test

Version 1 (no trend): $\Delta Y\_t = \alpha_0 + \alpha_1 Y\_\{t-1\} + \varepsilon\_t$

Version 2 (with trend): $\Delta Y\_t = \alpha_0 + \lambda t + \alpha_1 Y\_\{t-1\} + \varepsilon\_t$

H_0: $\alpha_1 = 0$ (unit root — NOT stationary) | H_1: $\alpha_1 < 0$ (stationary)

DF version	5% critical value
Version 1	**−2.86**
Version 2	**−3.41**

Reject H_0 if t-stat is **more negative** than critical value.

EXAM-DAY CHECKLIST

Step 1 — Stationarity (TS_C)

- ☐ Plot every variable against time
- ☐ Choose DF version (V1: no trend / V2: trend or unsure)
- ☐ Run DF on each variable, read t-stat on Y_{t−1}
- ☐ Compare to adapted critical values (NOT standard t-table)
- ☐ Unit root present? → take ΔY_t, re-test on differences

Step 2 — Fit the Model (TS_D)

- ☐ Model type given in question (static / DL(q) / AR(1))
- ☐ For DL: start at q_max, drop top lag while not significant
- ☐ Report all coefficients with standard errors and t-stats
- ☐ Calculate and report total multiplier (Σβ_k or β/(1−γ))
- ☐ Interpret: distinguish temporary vs. maintained; levels vs. log-returns

Step 3 — Assumption Check (TS_B)

- ☐ State H_0 and H_1 explicitly
- ☐ Write the auxiliary regression (all original X's + lagged residual)
- ☐ Compute LM = n_aux × R^2_aux
- ☐ Compare to $\chi^2(1) = 3.841$ (or use p-value)
- ☐ Write a full conclusion sentence

COMMON EXAM PITFALLS

1. "Reject H_0 means non-stationary" — WRONG

- For the **Dickey-Fuller** test: Reject H_0 means the series **IS stationary** (H_0 is the unit root).
- For the **LMSC** test: Reject H_0 means **autocorrelation IS present** (H_0 is no autocorrelation).
- These are opposite! Know which test you are running before concluding.

2. "Take differences whenever the series trends" — BE CAREFUL

- Only take differences if the DF test confirms a **stochastic trend** (unit root, $|\rho| = 1$).
- If the trend is **deterministic** ($|\rho| < 1$, $\lambda \neq 0$), add t as a regressor instead.
- **Over-differencing** creates artificial negative autocorrelation and loses information.

3. "Use Durbin–Watson for all models"

- DW is **invalid** whenever Y_{t−1} appears as a regressor (AR(1) models).
- Always use **LMSC** — it works for static, DL(q), and AR(1) models.

4. "Total multiplier = β_0 only"

- For DL(q): total multiplier = **$\beta_0 + \beta_1 + \dots +$ β_q** (sum of ALL lag coefficients).
- For AR(1): total multiplier = **$\beta/(1-\gamma)$**, not just β.
- β_0 alone is the **immediate effect** only.

5. "Unit changes when X is log-differenced"

- If you differenced log(X), a 1-unit rise in $\Delta\ln(X)$ means a 1 **percentage point** rise in the growth rate — not a 1-unit rise in the level.
- Interpret in terms of percentage changes.

6. "Skip the autocorrelation test to save time"

- The LMSC test is worth dedicated exam marks. Always report all five elements: H_0/H_1, the auxiliary regression, $LM = n \times R^2$, comparison to 3.841, and conclusion.

7. "Use the regular t-table for the Dickey-Fuller t-statistic"

- The DF t-statistic does NOT follow a t-distribution under H_0.
- Standard critical values (±1.96, ±2.58) do not apply.
- Always use the **adapted DF critical values**: −2.86 (V1, 5%), −3.41 (V2, 5%).

ONE-PAGE SUMMARY

```
FOUNDATIONS  →  Regression: Y = α + βX + ε. OLS finds best-fitting β.
                R² measures fit (0–1). t-statistics test if β ≠ 0.
           p-value < 0.05 → significant. Five classical assumptions must hold.

RECOGNISE      →   Time  series:  order  matters.  Define  lags  X_{t-
k} and differences ΔX_t.
           Watch sample size (n > 100 ideal). Use real monetary values.

STABILISE    →  Time plot + Dickey-Fuller on every variable.
              Deterministic trend (λ≠0, |ρ|<1)? Add t as regressor.
           Stochastic trend / unit root (|ρ|=1)? Take first differences.
           DF uses special critical values (−2.86 or −3.41), NOT ±1.96.
```

MODEL → Fit DL(q) or AR(1) on stationary variables.
DL(q): top-down selection of q; total multiplier = $\Sigma\beta_k$.
AR(1): total multiplier = $\beta/(1-\gamma)$; requires $|\gamma| < 1$.
Interpret: temporary vs. maintained, log-returns vs. units.

DIAGNOSE → LMSC test: auxiliary regression with lagged residual; $LM = n \cdot R^2 \sim \chi$
Reject H_0 → autocorrelation → re-specify (prefer DL over AR).
DW statistic: quick check only (not valid for AR models).

End of Complete Beginner's Guide to Time Series Regression